Step 8 Exploratory Back-ROI Item-Pair Report

Overview

Metric	Value
Included participant-sessions from Step 7	36
Concrete questions	15
Abstract questions	15
Tested concrete-abstract pairs	225
Pairs surviving FDR	0
Minimum paired N threshold	10
Matched-pair rule	—z(log word count difference)— ¡= 0.5 and —z(ENEM correctness difference)— ¡= 0.5
Selected-pair follow-up rule	All FDR-significant pairs, otherwise the top 3 lowest-q descriptive pairs

Exploratory status. Step 8 was implemented exactly as an exploratory item-pair deviation from the main pipeline, using the frozen Step 7 cohort and the Step 7 duration-aware back-ROI outcome.

Top pair summary. Top pair: 2018_CH_77 versus 2021_CH_86 (mean difference=1.864e-07, raw p=0.003431, q=0.596248).

Final conclusion. The exploratory item-pair analysis did not identify any concrete-abstract question pairs with reliable back-ROI differences after Benjamini-Hochberg correction. This supports the view that the back ROI does not show robust item-specific discrimination in the current dataset. The lowest-q pair was $2018_CH_77versus2021_CH_86(meandifference = 1.864e-07, rawp = 0.003431, q = 0.596248).Nomatched-pairsubsetcomparisonsurvivedFDRcorrectionunderthepredeclaredmatchi$

Top Pairwise Results

Concrete	Abstract	N	Mean diff	Raw p	FDR q	d_z
2018_CH_77	2021_CH_86	36	1.864e-07	0.003431	0.596248	0.523
2020_CH_59	2021_CH_86	36	1.766e-07	0.006870	0.596248	0.479
2018_CH_77	2019_CH_64	36	1.879e-07	0.007950	0.596248	0.469
2020_CH_59	2019_CH_64	36	1.781e-07	0.016495	0.673295	0.420
2021_LC_26	2019_CH_56	35	-1.368e-07	0.017195	0.673295	-0.423
2020_CH_59	2021_CH_83	36	1.253e-07	0.017955	0.673295	0.414
2020_CH_84	2019_CH_64	35	1.222e-07	0.028812	0.853138	0.386
2018_CH_77	2021_CH_83	36	1.351e-07	0.038986	0.853138	0.357
2017_CH_69	2021_CH_86	36	1.251e-07	0.039542	0.853138	0.356
2020_CH_59	2022_CH_71	36	1.066e-07	0.039576	0.853138	0.356
2017_CH_47	2021_CH_86	36	1.080e-07	0.046574	0.853138	0.344
2020_CH_78	2021_CH_86	35	1.287e-07	0.058036	0.853138	0.332

Top Question Rankings

Rank	Question	Cond.	Centered score (uM)	N
1	2018_CH_77	Concrete	0.076	36
2	2019_CH_56	Abstract	0.072	36
3	2020_CH_59	Concrete	0.066	36
4	2021_CH_65	Abstract	0.049	36
5	2021_CH_57	Abstract	0.040	36
6	2021_CH_51	Abstract	0.035	36
7	2022_CH_51	Abstract	0.034	36
8	2022_CH_60	Abstract	0.029	36
9	2020_CH_77	Abstract	0.027	35
10	2020_CH_76	Abstract	0.019	36
11	2019_CH_65	Abstract	0.017	36
12	2020_CH_78	Concrete	0.016	35

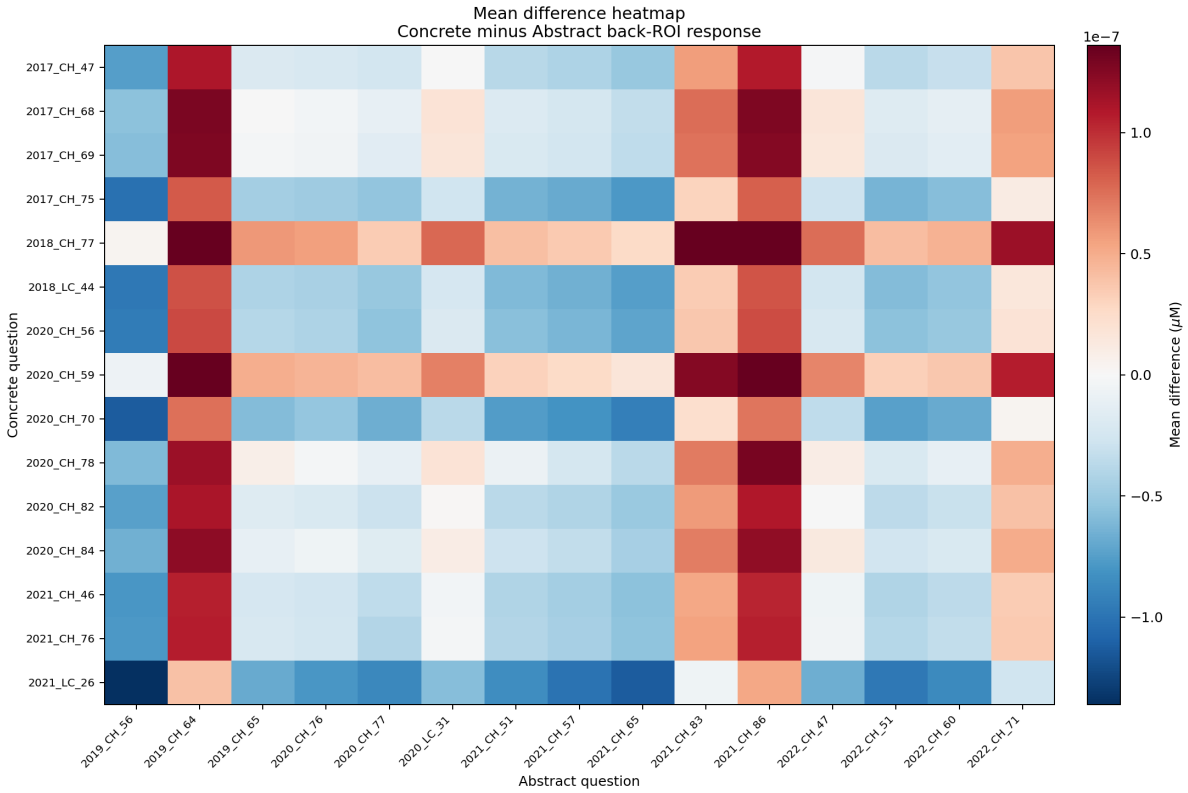

Figure 1: Mean back-ROI concrete-minus-abstract difference heatmap for all tested item pairs.

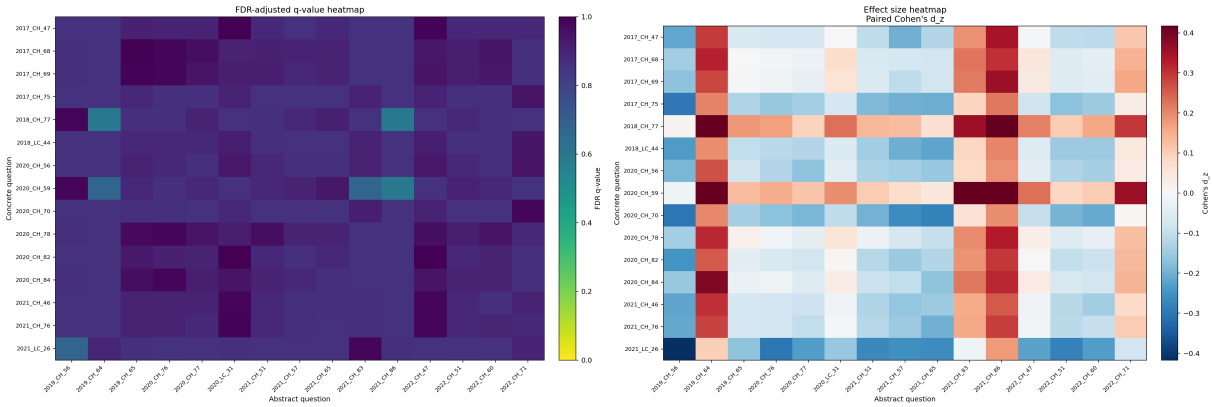

Figure 2: FDR-adjusted q-values and paired effect sizes for all tested item pairs.

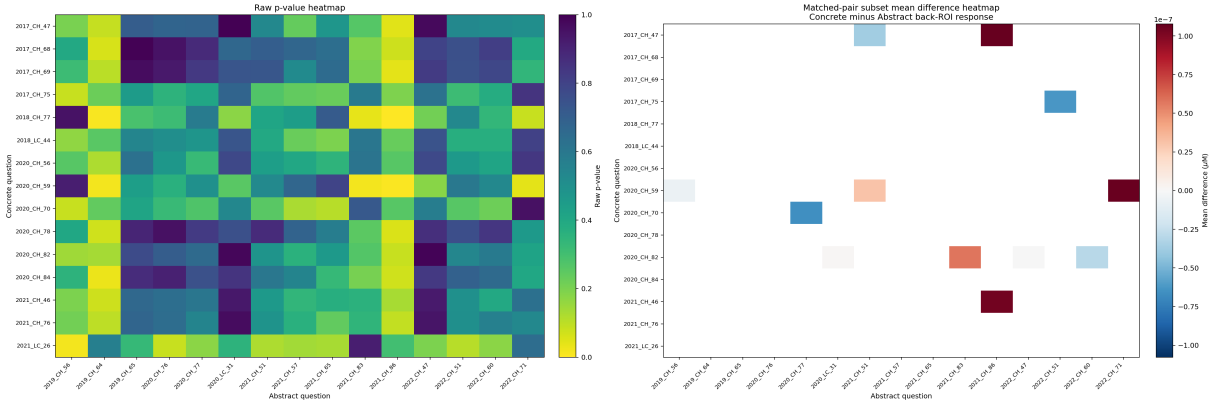

Figure 3: Raw p-values for all tested item pairs and mean differences restricted to the matched-pair robustness subset.

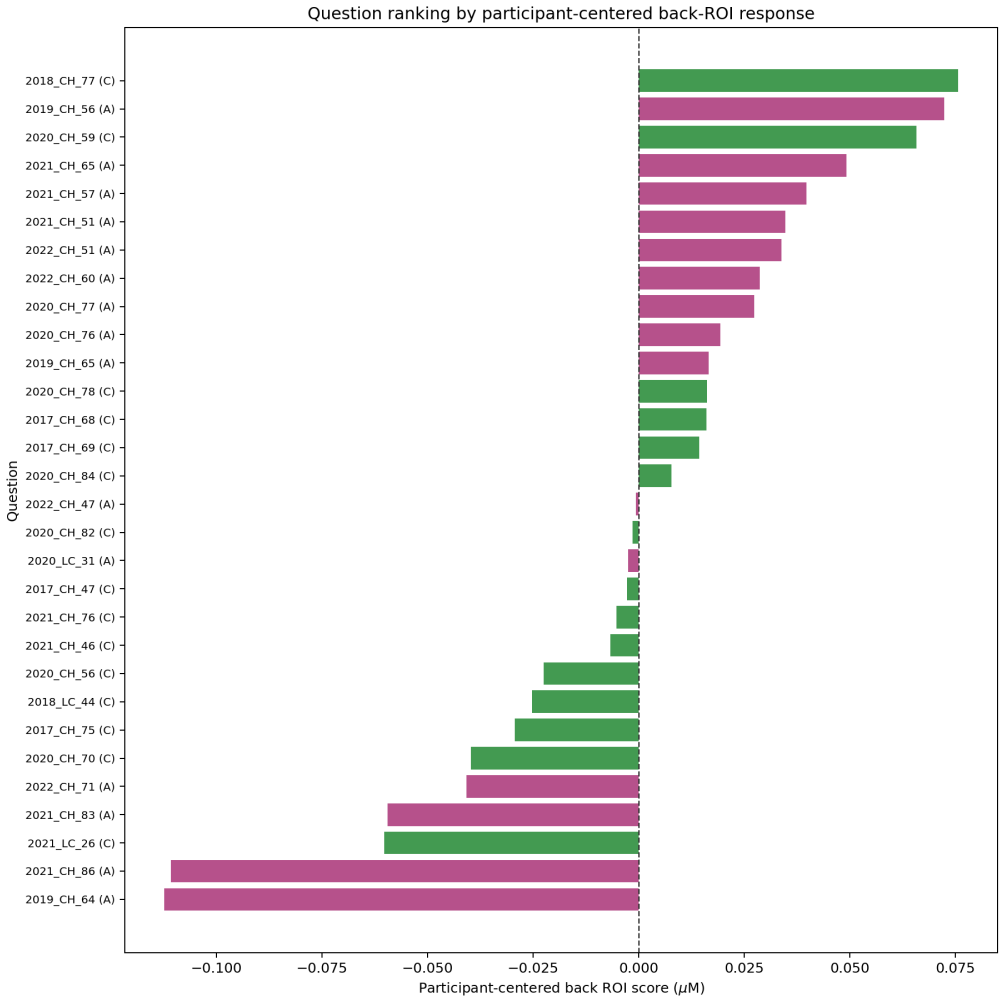

Figure 4: Question ranking by participant-centered back-ROI response.

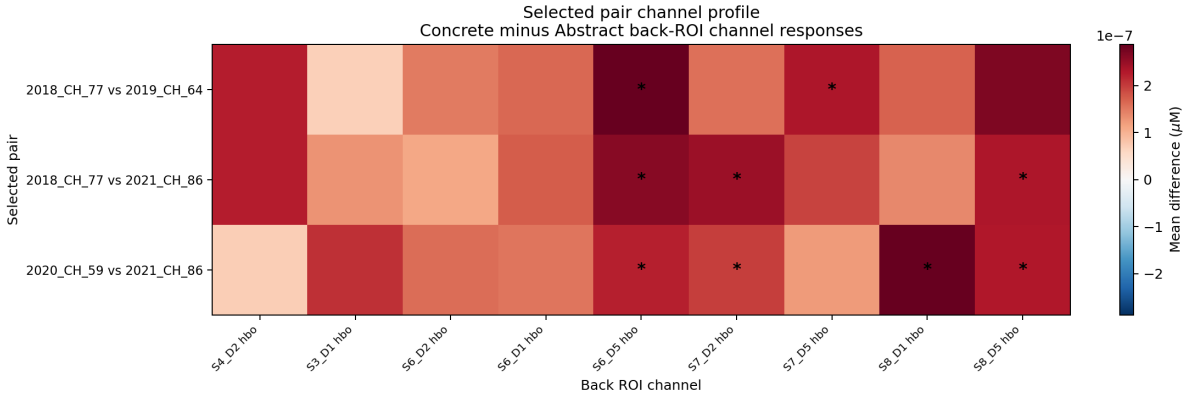

Figure 5: Optional channel-level follow-up for the selected item pairs. Asterisks mark channels surviving within-pair FDR correction.